

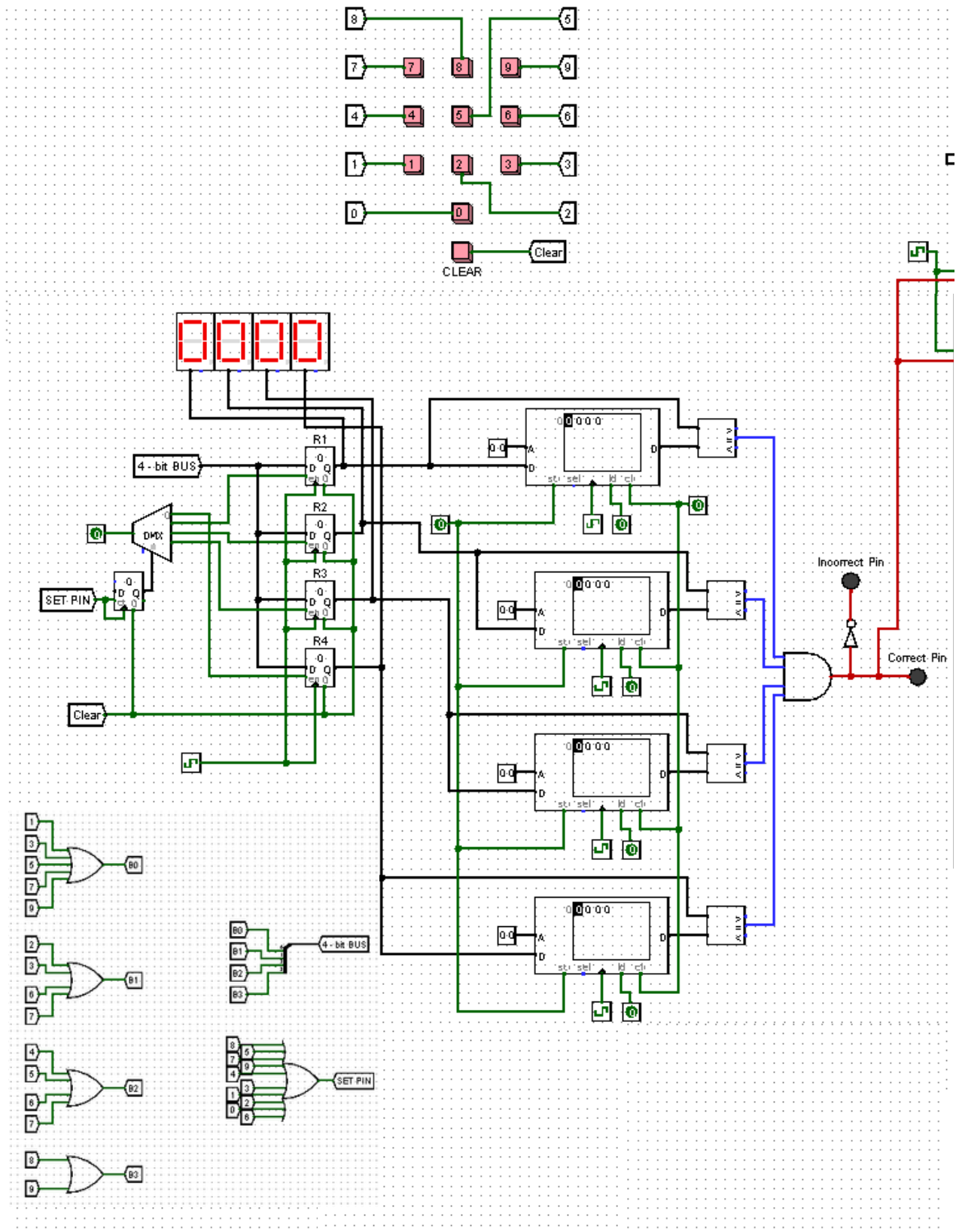
DIGITAL ELECTRONICS

ATM SIMULATION CIRCUIT

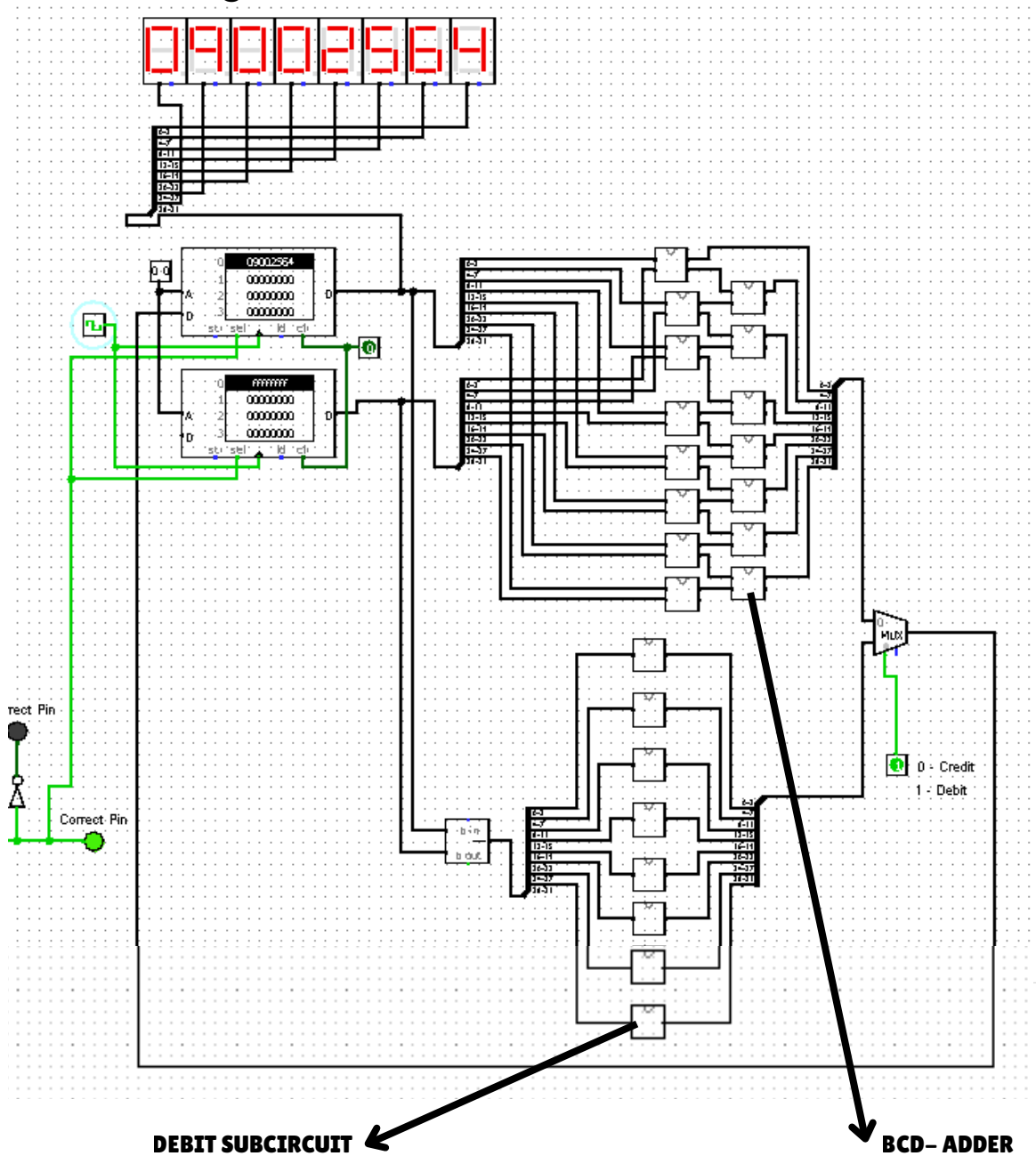
- The project simulates an ATM system using digital logic circuits in Logisim software.
- It is designed to verify a 4-digit PIN entered by the user and compare it with a stored PIN value.
- The circuit uses registers to store both the set PIN and the entered PIN.
- Comparators and logic gates (AND, OR, NOT) are used to check whether the two PINs match.
- If the entered PIN matches the stored PIN, a "Correct PIN" indicator (LED/output) turns ON.
- If the PIN is incorrect, an "Incorrect PIN" indicator lights up instead.
- The system includes a keypad-like input for entering the PIN and a Clear button to reset inputs.
- MUX and DEMUX components are used to control data flow between different parts of the circuit.
- A BCD Adder subcircuit is used for arithmetic operations (useful in balance-related or transaction extensions).
- The design combines combinational and sequential logic to simulate real ATM operations like PIN verification.
- This project demonstrates practical understanding of digital electronics concepts, including data storage, logical control, and signal flow.

CIRCUIT

- The circuit is divided into two main logical sections:
 - Left Block (**PIN verification**): Contains a 4-bit bus, a set of 4-bit registers, and a multiplexer (MUX) used to load an initial or custom value into a small 4-digit register array. This section also manages the global Clear and SET PIN control.

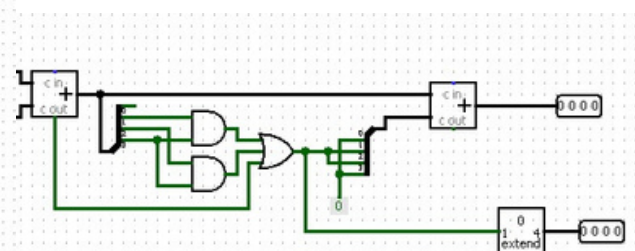
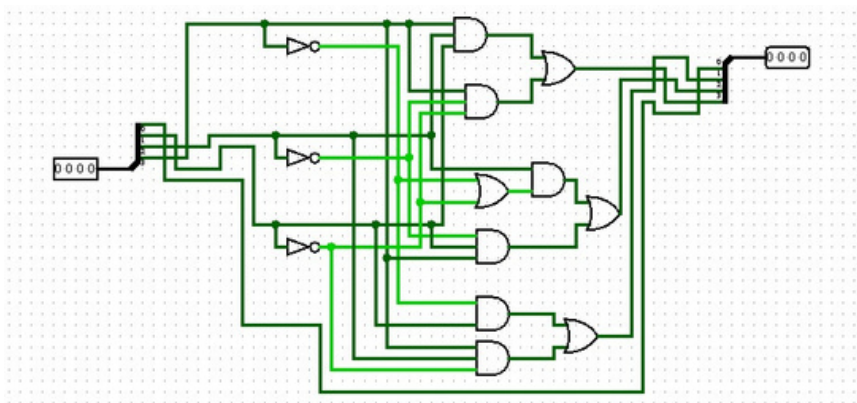


- Right Block (Accumulator/**Transaction Logic**): This is the core of the system. It features two larger RAM (likely 32-bit each, cascaded) to store a large 8-digit value. A complex array of gates and MUXes performs the Credit/Debit logic based on the 0: Credit or 1: Debit control signal.



DEBIT SUBCIRCUIT

BCD- ADDER



WORKING

- Clock and Clear:
 - A Clock pulse synchronizes all operations. Registers only update on the active edge of the clock signal.
 - The CLEAR pin resets all registers and sets the displayed count/balance to zero.
- Value Loading (Left Block):
 - This section allows an initial value to be loaded. The 4-bit BUS feeds a new value into a registers.
 - The SET PIN acts as a control signal for the MUX. When the SET PIN is active, the MUX selects the input from the bus, allowing a new value to be loaded into the registers on the next clock pulse, overriding the current count/balance.
- Credit/Debit Transaction Logic (Right Block):
 - The main update mechanism is governed by a control pin connected to a MUX (labeled 0: Credit, 1: Debit).
 - Credit (Increment): When the control pin is set to '0' (Credit), the logic calculates the value of the current balance plus a defined transaction amount (likely a fixed value or a variable input not fully shown). The result is fed back to the registers.

- Debit (Decrement): When the control pin is set to '1' (Debit), the logic calculates the value of the current balance minus a defined transaction amount.
- The complex network of gates (Adders/Subtractors) calculates the next state based on the current stored value and the transaction type (Credit/Debit).
- Display Output:
 - The current value stored in the primary (8-digit) registers is continuously output to the large 7-segment display. This display shows the updated count/balance after each successful transaction and clock cycle (e.g., 09002564 or 09347856).

COMPONENTS USED

COMPONENT NAME	TYPE	PURPOSE
BUTTON / PINS	INPUT/OUTPUT	REPRESENTS NUMBER(0-9)AND CLEAR
TUNNEL	WIRING	USED FOR MAKING THE CIRCUIT VISUALLY CLEAR
CLOCK PULSE	INPUT TIMING SOURCE	SYNCHRONIZES THE LOADING AND UPDATING OF ALL SEQUENTIAL ELEMENTS
LOGIC GATES	COMBINATIONAL DEVICES	IMPLEMENTS ARITHMETIC LOGICS
REGISTER (4BIT)	SEQUENTIAL ELEMENTS	USED TO STORE THE STORE THE CURRENT PIN / BALANCE
COUNTER	SEQUENTIAL ELEMENT	USED FOR GENERATION OF SEQUENTIAL OUTPUTS
MULTIPLEXER	COMBINATIONAL DEVICES	USED TO SELECT ONE VALUE AMONG THE PROVIDED INPUTS
RAM	MEMORY DEVICES	USED TO STORE THE PIN SET BY THE USER AND ALSO INPUT BALANCE AND AMOUNT(DEBIT/CREDIT) BY THE USER
HEX DISPLAY	OUTPUT INDICATOR	DISPLAYS THE CURRENT BALANCE AND PIN

CONCLUSION

- The DIGITAL ATM SIMULATOR project successfully applies fundamental principles of Digital Electronics to model a practical data processing system. The circuit effectively combines sequential memory (Registers & Counter) with combinational arithmetic logic (ALU and MUXes) to create a reliable and secure accumulator. This implementation serves as an excellent illustration of how digital circuits can manage state, perform controlled transactions, and enforce security access, thereby enhancing the understanding of real-time problem-solving in digital systems.